

7 (short) Before the following PIC12F629 code is executed, the contents of memory locations 0x30 and 0x31 are the decimal numbers 20 and 50 respectively:

```
        movlw 0x31;
        movwf FSR;
        call sr;
        decf FSR;
        call sr;
        sleep;
.....
sr      rrf INDF;
        movlw 0x10;
        addwf INDF;
        return;
```

- (a) Explain the function of each step; [5]
- (b) What are the contents of W and memory locations 0x30 and 0x31 after the code has executed? [3]
- (c) Determine the number of clock cycles and time taken to execute the code assuming a 20 MHz clock. [2]